

YES. Ab tumhara raw idea actually kaafi strong hai. Aur honestly, tumhare diye hue 5 lab ideas mein se mujhe ye idea zyada “real ML system” banne ki potential rakhta hai.

Lekin ek important correction hai:

“Ye banda loan pay kar sakta hai ya nahi?” → Classification

“Is bande ko maximum kitna loan dena chahiye?” → Risk-based loan amount recommendation / constrained regression

Dono ko combine karoge to project kaafi impressive banega.

💰 Project concept: AI Loan Eligibility + Maximum Affordable Loan

Imagine ek person loan ke liye apply karta hai.

System sirf Yes/No nahi boleگا.

It will say:

Eligibility: Approved

Default Risk: 8.3%

Recommended Maximum Loan: Rs. 650,000

Suggested Tenure: 24 months

Estimated Monthly Installment: Rs. 32,400

Risk Level: Low

Main reasons: Stable income, low existing debt, good repayment history

Ya:

✗ Not recommended

Default risk: 71%

Main issue: high debt-to-income ratio + unstable employment + previous delinquencies.

Ab ye actual product lagta hai.

☐ Sabse pehle: kaun-kaun se factors hone chahiye?

Main inko categories mein divide karunga.

1. Personal/Demographic

Possible features:

- Age
- Marital status
- Number of dependents
- City/region
- Education level
- Household size

Lekin: protected/sensitive characteristics ko automated credit decision mein blindly use nahi karna chahiye. For a university prototype, kuch demographic fields ko either exclude karna ya fairness analysis ke liye separately evaluate karna better hai.

2. Income — VERY IMPORTANT

Ye sabse important category hogi.

- Monthly income
- Annual income
- Salary vs self-employed
- Primary income
- Secondary income
- Other recurring income
- Income variability
- Income growth/stability
- Months/years of employment

Example:

Person A:

Rs. 150k/month, stable for 5 years

Person B:

Rs. 150k/month, income highly variable

Same income $\neq$ same repayment capacity.

3. Employment

Very useful:

- Employment status
- Employment type
- Years at current job
- Total work experience
- Employer stability
- Permanent/contract
- Self-employed/business
- Business age

For a Pakistani version:

Government
Private
Business
Freelancer
Daily wage
Other

But these categories need careful treatment because they can create bias.

4. Financial Obligations

This is where your “maximum loan” idea gets interesting.

Current obligations:

- Existing loan amount
- Existing monthly installments
- Credit-card outstanding
- Other debts
- Rent
- Household expenses
- Education expenses
- Other recurring obligations

Then calculate:

Debt-to-Income Ratio

$$DTI = \text{Monthly Debt} / \text{Monthly Income}$$

Example:

Income = Rs. 200k

Existing debt payments = Rs. 70k

DTI = 35%

Then new loan ki installment add karke dekho:

Can this person realistically afford another Rs. 30k/month?

5. Credit History

This is **gold** for the project.

Features:

- Credit history length
- Number of previous loans
- Number of closed loans
- Number of active loans
- Previous defaults
- Late payments
- Days past due
- Number of missed installments
- Previous loan repayment ratio
- Credit utilization
- Number of credit accounts
- Recent credit inquiries

For example:

Previous loans = 5
Successfully repaid = 5
Late payments = 0
Default = 0

vs.

Previous loans = 5
Successfully repaid = 2
Late payments = 8
Default = 2

These applicants shouldn't receive the same risk score.

6. 💰 Proposed Loan Information

Applicant jo **new loan** maang raha hai uske features:

- Requested loan amount
- Loan purpose
- Loan tenure
- Interest rate
- Secured/unsecured
- Installment frequency
- Down payment, if applicable

Example:

Income = Rs. 150k
Requested loan = Rs. 5 million

System should say:

✗ Requested amount exceeds estimated repayment capacity.

But maybe:

Recommended maximum = Rs. 1.8 million

7. 🏠 Derived Features — THIS is where your ML project gets good

Raw data se directly model train mat karna.

Feature engineering karo.

Debt-to-income

$$DTI = \frac{\text{Existing Monthly Debt}}{\text{Monthly Income}}$$

Loan-to-income

$$LTI = \frac{\text{Requested Loan}}{\text{Annual Income}}$$

Installment-to-income

$$ITI = \frac{\text{New Monthly Installment}}{\text{Monthly Income}}$$

Credit utilization

$$\text{Utilization} = \frac{\text{Outstanding Credit}}{\text{Total Credit Limit}}$$

Repayment success rate

$$\text{RepaymentRate} = \frac{\text{Successfully Repaid Loans}}{\text{Total Previous Loans}}$$

Employment stability

$$\text{EmploymentStability} = f(\text{JobYears}, \text{IncomeVariation})$$

These engineered features will make your project much more meaningful than simply throwing 15 columns into Random Forest.

🔑 Now the REAL ML problem

Main is project ko 3-stage system banaunga.

Model 1 — Default Probability

Question:

Will the applicant default?

Target:

0 = Repays
1 = Defaults

Models:

- Logistic Regression
- Random Forest
- SVM
- Naive Bayes

Then compare them.

Metrics:

- ROC-AUC
- Precision
- Recall
- F1
- PR-AUC
- Confusion Matrix

This directly matches your lab's classification requirements.

Model 2 — Maximum Affordable Loan

Yahan tumhara original idea aata hai.

Question:

Given this applicant's financial profile, what loan amount is reasonably affordable?

But **important**: dataset mein "maximum loan they can afford" usually directly labelled target nahi hota.

So you have two options.

Option A — Regression

Historical approved loans se learn:

Applicant profile → loan amount associated with successful repayment

But this can accidentally learn **what lenders historically approved**, not the true maximum affordable amount.

Better option B — Financial constraint + ML

ML predicts:

Probability of default at a proposed loan amount

Then system tests different amounts.

For example:

Rs. 300k → 4% risk
Rs. 500k → 6% risk
Rs. 700k → 11% risk
Rs. 900k → 19% risk
Rs. 1.2m → 35% risk

Then define a transparent project threshold, e.g.:

acceptable predicted risk $\leq 10\%$

So:

Recommended maximum $\approx$ Rs. 700k

THIS is much more interesting than ordinary regression.

☐ **And Model 3 can be Anomaly Detection**

Suppose someone has:

Income = Rs. 100k
Loan requested = Rs. 10 million
Reported debt = Rs. 0

Even if classifier says something weird because of limited information, anomaly detector can flag:

🔗 Financial profile is unusual

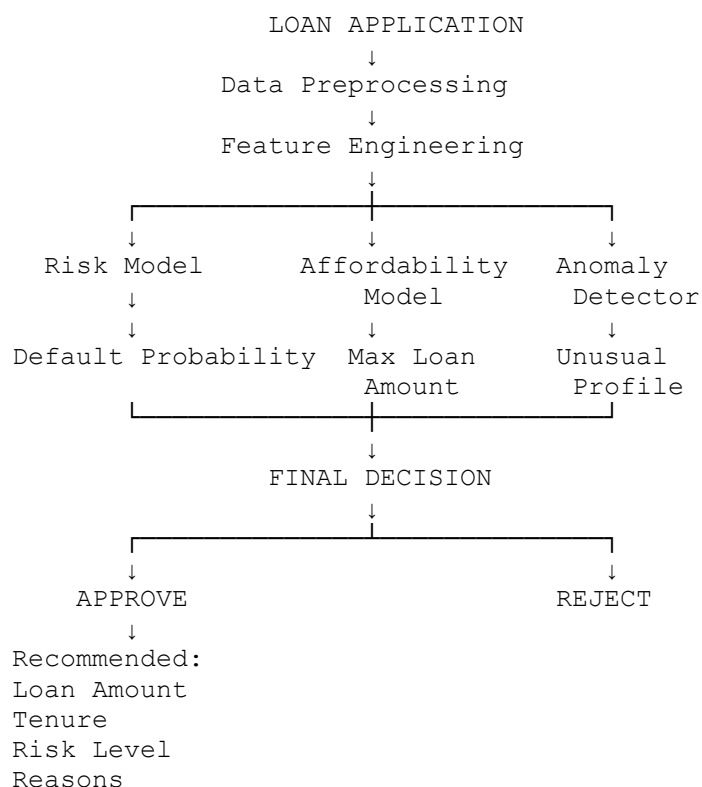
Use:

- Isolation Forest
- K-Means
- Hierarchical Clustering

Then your project satisfies **supervised + unsupervised ML**.

□ Complete system

Tumhara final architecture something like this:



PK Ab sabse important question: Pakistan ka data milega?

Exact Pakistani bank loan dataset?

This is the difficult part.

Banks ke real applicant + repayment records generally public nahi hote because they're sensitive financial data.

So don't design your project assuming you'll get HBL/UBL/Meezan/NBP's real customer records.

BUT...

Public international datasets **excellent** hain.

For example, the Kaggle **Credit Risk Dataset** is CC0/public domain and has around 32k records with variables such as age, annual income, home ownership, employment length, loan intent, loan grade, loan amount, interest rate, loan status/default, loan-to-income, historical default and credit-history length. ([Kaggle](#))

Even better, **Home Credit Default Risk** provides a much richer ecosystem: applications, previous applications, credit-bureau records, monthly balances, previous loans, credit-card balances and installment-payment histories. ([Kaggle](#))

Another public credit-risk dataset includes credit score, employment years, annual income, monthly debt, credit history, open accounts, credit problems, current credit balance, bankruptcies and tax liens. ([Kaggle](#))

So data availability is NOT a blocker.

PK Lekin main isko Pakistani kaise banaunga?

Ye important hai.

Dataset foreign hai → koi issue nahi.

Problem framing Pakistani rakho.

For example, UI:

AI Loan Assessment — Pakistan

Applicant enters:

Monthly Income: Rs. 180,000
Employment: Private

Employment Duration: 4 years
Existing Monthly Debt: Rs. 35,000
Requested Loan: Rs. 1,000,000
Tenure: 24 months
Previous Defaults: 0

System:

LOAN ASSESSMENT

Default Risk	7.4%
Risk Level	LOW ☐
Requested Amount	Rs. 1,000,000
Recommended Max	Rs. 1,150,000
Estimated EMI	Rs. XX,XXX
Decision	APPROVE

Main Factors:

- ✓ Stable employment
- ✓ Low debt burden
- ✓ Good repayment history
- ⚠ Moderate loan-to-income ratio

Data foreign olabilir, but system Pakistan-oriented prototype ho sakta hai.

⚠ Ek aur important cheez: "maximum loan" ko carefully define karna

Main project ka naam:

✗ "Loan Approval Prediction System"

nahi rakhunga.

Bahut generic hai.

Instead:

“Explainable AI-Based Loan Repayment Risk & Affordable Credit Limit Estimation”

Ya simpler:

“AI-Based Loan Eligibility and Maximum Affordable Loan Prediction”

Aur claim ye nahi hoga:

“AI decides who deserves a loan.”

Rather:

“AI estimates repayment risk and proposes a financially constrained loan amount to assist a human credit decision-maker.”

Ye academically aur ethically dono better hai.

Isko aur unique banane ka ek killer feature

What-if simulator.

Applicant ko sliders do:

Monthly Income	Rs. 150k	————●————	Rs. 300k
Existing Debt	Rs. 50k	——●————	Rs. 100k
Loan Amount	Rs. 500k	————●————	Rs. 2m
Tenure	12	————●————	60 months

Aur live:

Risk: 8% → 14% → 22%

Then:

“If requested loan decreases from Rs. 1.5M to Rs. 1.0M, estimated risk falls from 18% to 9%.”

That would make your project feel like an actual fintech product rather than a university ML assignment.

★ Final verdict

Tumhara raw idea **discard karne wala nahi — actually develop karne wala hai.**

I'd rate it:

Real-world problem: 10/10

Data availability: 9/10

ML concepts: 10/10

Feature engineering: 10/10

Demo potential: 9/10

Uniqueness: 8.5/10 → **10/10 with what-if + max-affordable-loan feature**

Aur sabse achi baat: tumhari lab ke requirements naturally fit hote hain:

Preprocessing → Classification → Regression → Ensemble → Clustering/Anomaly Detection → Explainability → Business decision.

Agar tum isko choose karti ho, **next step mein main exact dataset(s) shortlist karke bata sakta hoon ke kaunsa use karo, usmein kaunse columns hain, kaunse columns missing hain, aur tumhare “maximum affordable loan” target ko mathematically kaise define karna hai**—taake project baad mein target-variable ke issue mein phans na jaye.